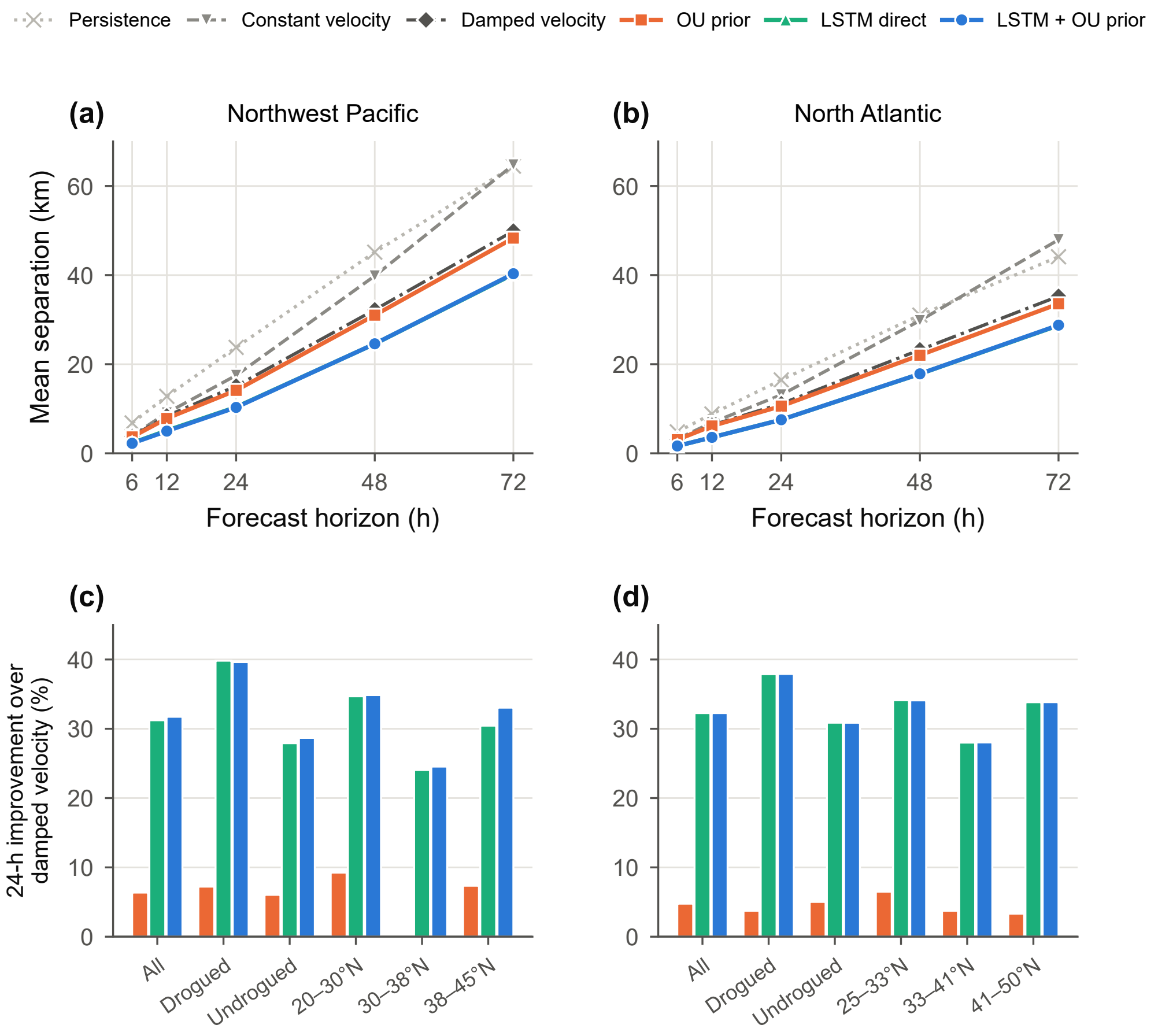

Persistence
Constant velocity
Damped velocity
OU prior
LSTM direct
LSTM + OU prior
(a)
Northwest Pacific
(b)
North Atlantic
Mean separation (km)
0
20
40
60
6
12
24
48
72
Forecast horizon (h)
(c)
(d)
24-h improvement over damped velocity (%)
0
10
20
30
40
All
Drogued
Undrogued
20–30°N
30–38°N
38–45°N
25–33°N
33–41°N
41–50°N